

Mexora RH Intelligence

IT Job Market Analysis — Morocco

Data Engineering Mini-Project 2

Final Analytical Report

Prepared for the HR Director, Mexora — Tanger, Morocco

5,000

IT Job Offers

47,560

Skill Extractions

23

Months of Data

13

Cities Analysed

3

Job Platforms

40

Companies Tracked

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Author: Matthews Lungu and Adjii Charles

Course: Data Engineering — End-of-Study Project

Stack: Python 3.12 · DuckDB · Parquet · Power BI

Repository: github.com/Matthews-Lungu/Mexora_rh_Data_lake

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1. Executive Summary

This report presents the findings of a data-driven market intelligence study conducted for Mexora, an e-commerce company based in Tanger, Morocco. The HR Director plans to recruit five data profiles to support the company’s digital growth. Rather than relying on intuition or isolated job postings, this study analysed **5,000 IT job offers** collected from the three main Moroccan recruitment platforms — Rekrute, MarocAnnonce, and LinkedIn Morocco — covering January 2023 to November 2024.

The analysis was conducted using a purpose-built **Bronze/Silver/Gold Data Lake**, processing raw job offer data through cleaning, NLP-based skill extraction, and DuckDB-powered aggregation. Five analytical questions were answered, producing clear, actionable recommendations for the HR Director.

Key Findings at a Glance

- **SQL is the #1 skill** demanded in Morocco, appearing in 58.6% of all IT job offers
- **Tanger ranks 3rd nationally** with 597 IT offers — a growing, less competitive market
- **Data Engineer median salary in Tanger: 19,000 MAD/month** (95 offers, statistically solid)
- **Tanger offers the highest remote work rate** (69.5% for Data Engineer) of all major cities
- **No salary premium for experience** — the Moroccan market pays similar salaries regardless of seniority, giving Mexora flexibility to hire both junior and senior profiles at competitive rates
- **Tanger Free Zone IT** (17,763 MAD avg) is Mexora’s strongest local competitor for data talent

2. Data Lake Architecture

The study is powered by a three-zone Data Lake built entirely in Python, using Parquet for analytical storage and DuckDB as the query engine. Figure 1 shows the complete data flow from raw sources to business consumers.

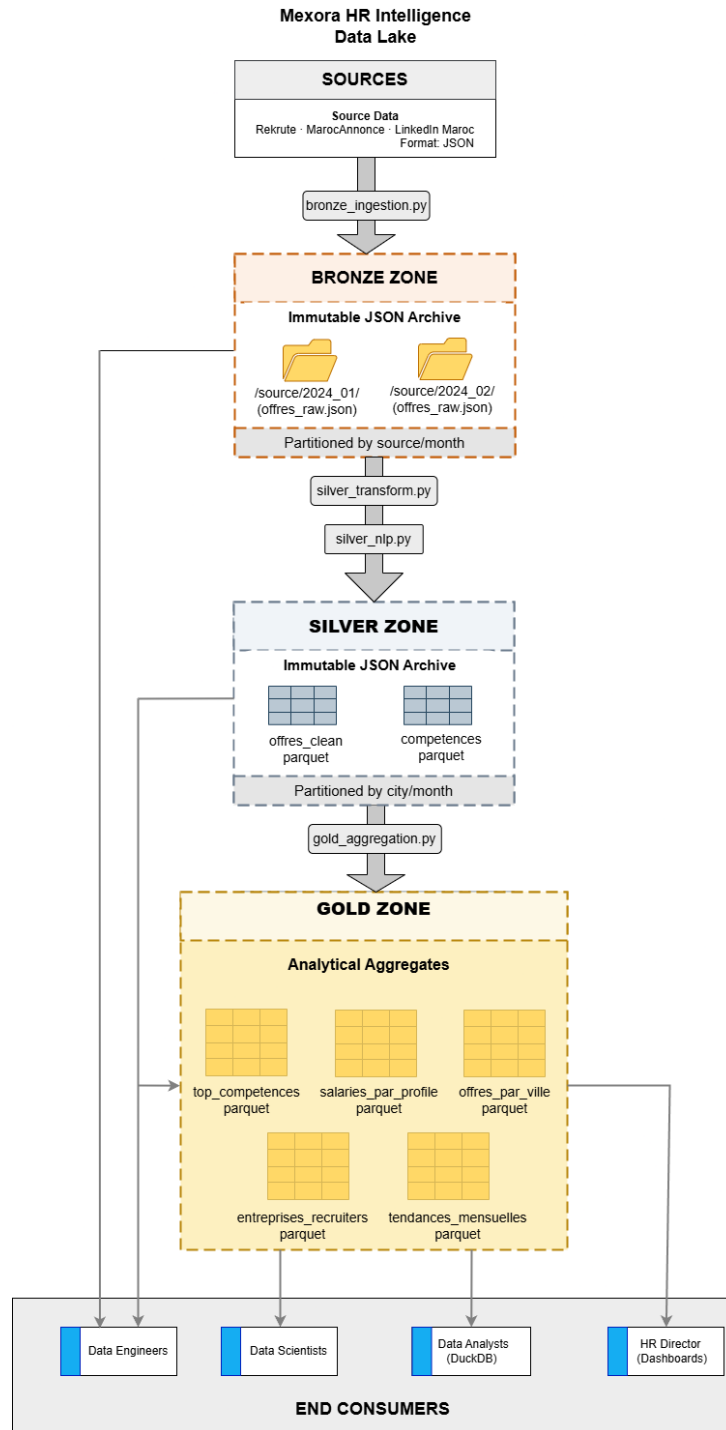


Figure 1. Bronze/Silver/Gold Data Lake architecture — Mexora RH Intelligence. Data flows from three job platforms (Rekrute, MarocAnnonce, LinkedIn) through three quality zones to end being utilized by their respective end consumers

2.1. Bronze Zone — Raw Immutable Archive

All 5,000 job offers are stored exactly as received, partitioned by source and month into 69 partition files. No field is ever modified in Bronze. This ensures full data lineage and allows the pipeline to be reprocessed from scratch at any time.

2.2. Silver Zone — Cleaned and Standardised

The Silver transformation layer applies six normalisation operations: job title classification into 11 canonical profiles, city name standardisation (50+ dirty variants mapped to 13 standard cities), salary parsing across 7 different formats, experience parsing, date validation, and NLP-based skill extraction (47,560 matches across 64 skills). **75.4% of offers are fully clean** with zero anomalies.

2.3. Gold Zone — Business Aggregations

Five Parquet tables are produced by DuckDB SQL on Silver data, each directly answering one of the five analytical questions. Total size of all five Gold tables: 39 KB — compact, fast, and dashboard-ready.

Pipeline Step	Output	Time	Quality
Bronze Ingestion	69 partitions, 5,000 offers	2.8s	0 anomalies
Silver Transform	5,000 rows × 33 columns	2.8s	75.4% clean
NLP Extraction	47,560 skill matches	74.1s	100% coverage
Gold Aggregation	5 analytical tables	2.2s	Query-ready
Total	Full pipeline	81.9s	

3. Market Analysis — Five Analytical Questions

3.1. Question 1 — What IT Skills Are Most in Demand in Morocco?

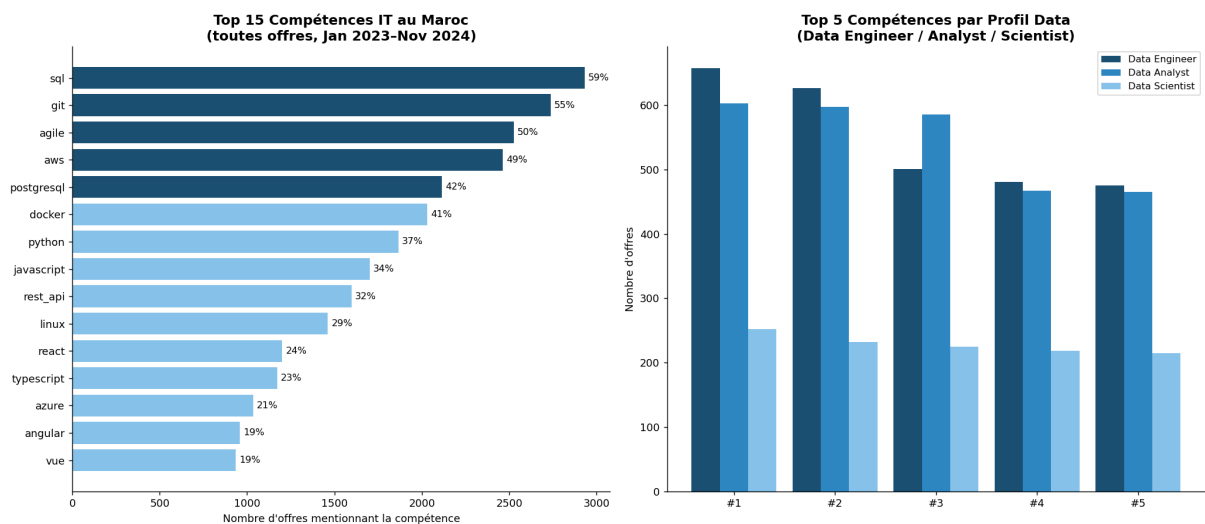


Figure 2. Left: Top 15 IT skills across all 5,000 job offers (Jan 2023–Nov 2024). Right: Top 5 skills per data profile — Data Engineer, Data Analyst, Data Scientist. Source: NLP extraction from Silver zone, aggregated in Gold `top_competences.parquet`.

SQL is the most demanded skill, appearing in 58.6% of all IT job offers, followed by Git (54.7%), Agile (50.5%), and AWS (49.3%). This confirms that foundational skills — relational databases, version control, and agile methodology — are non-negotiable regardless of the specific profile.

Among data profiles, significant divergences emerge. The **Data Engineer** is distinguished by strong demand for Spark (88.9% of DE offers), ETL (67.7%), and dbt (63.0%) — specialised pipeline tools absent from other profile top lists. The **Data Analyst** remains centred on SQL and BI tools (Power BI, Looker, Tableau), while the **Data Scientist** requires Python and ML frameworks (scikit-learn, TensorFlow, PyTorch).

When writing job descriptions, require SQL and Python as a mandatory baseline for all data hires. Differentiate profiles by adding Spark/dbt for the Data Engineer role, Power BI for the Data Analyst, and scikit-learn/PyTorch for the Data Scientist. Cloud skills (AWS or Azure) are expected across all three profiles.

3.2. Question 2 — Tanger vs. Casablanca vs. Rabat: Where Are the IT Opportunities?

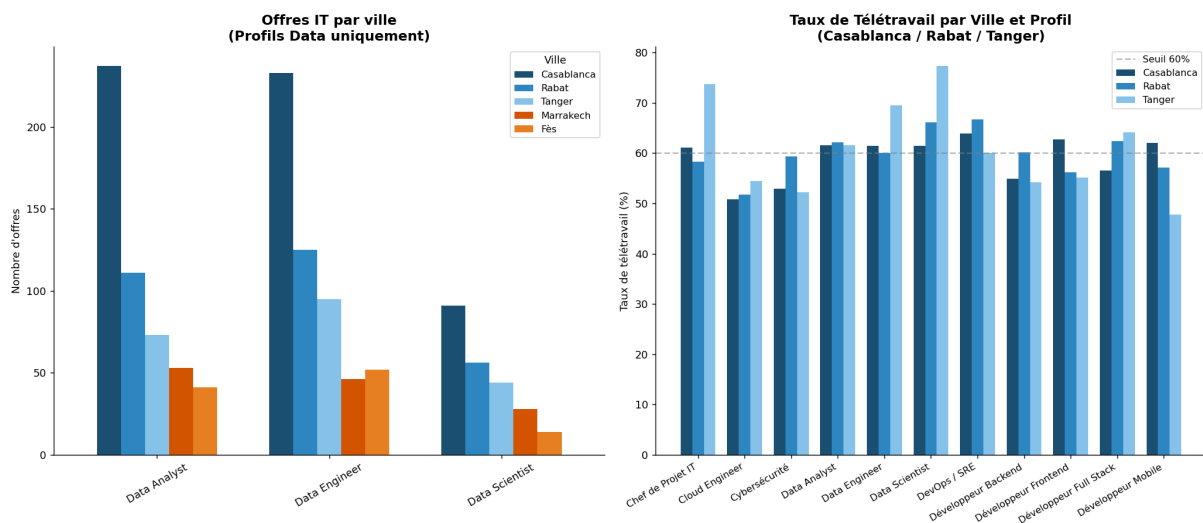


Figure 3. Left: Number of IT job offers by city for three data profiles. Right: Remote work rate (%) by city and profile for Casablanca, Rabat, and Tanger. Source: `offres_par_ville.parquet` (Gold zone).

Casablanca concentrates the majority of IT offers with 1,704 offers (34.1%), confirming its status as Morocco’s digital economic capital. Rabat follows with 900 offers (18.0%), driven by public sector institutions. **Tanger ranks third with 597 offers (11.9%)** — a strong result for a city considered emerging in Morocco’s IT ecosystem.

The most actionable finding for Mexora is the remote work comparison. **Tanger leads all major cities with a 69.5% remote rate for Data Engineer**, compared to 61.4% in Casablanca and 60.0% in Rabat. Candidates in Tanger are both accustomed to and specifically seeking flexible working arrangements. Tanger also leads for Data Scientist (77.3%) and Chef de Projet IT (73.7%).

Mexora's Tanger location is an asset, not a handicap. The local market is active but less competitive than Casablanca. By offering hybrid working arrangements (at least 2–3 days remote per week), Mexora can attract candidates who refuse to relocate to Casablanca, including profiles currently working remotely for Casablanca-based employers.

3.3. Question 3 — What Are the Salary Ranges by IT Profile in Morocco?

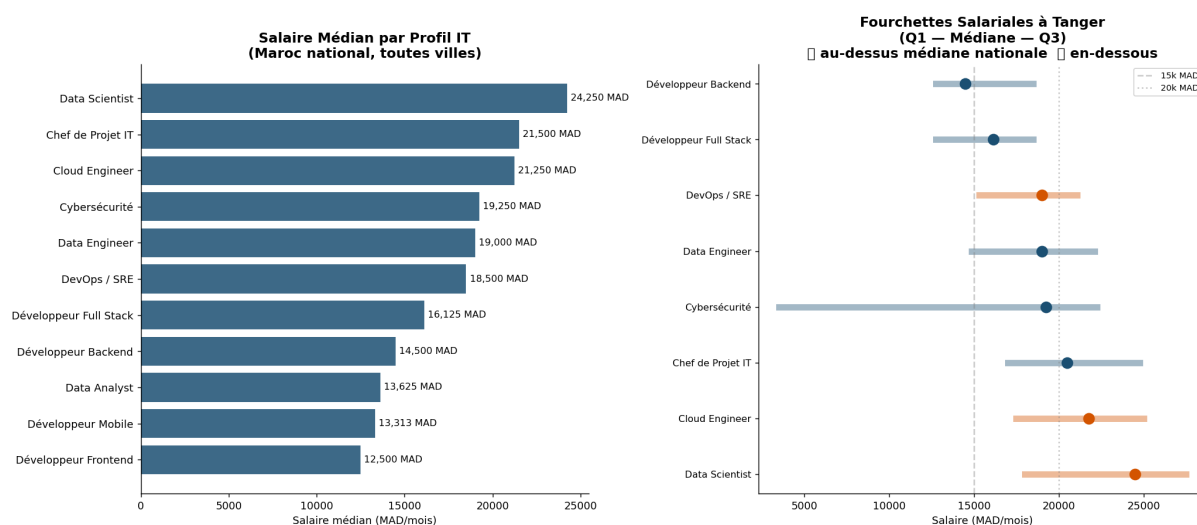


Figure 4. Left: National median salary by IT profile (all cities, 2023–2024). Right: Salary ranges at Tanger — Q1, median (dot), and Q3 for each profile. Orange = above national median. Blue = below or equal to national median. Source: `salaires_par_profil.parquet` (Gold zone).

Salaries in Moroccan IT follow a clear hierarchy. The **Data Scientist** commands the highest median at **24,250 MAD/month** nationally, followed by Chef de Projet IT (21,500 MAD) and Cloud Engineer (21,250 MAD). The **Data Engineer** sits at **19,000 MAD/month** nationally — exactly aligned with its Tanger median.

At Tanger specifically, salary distributions are healthy and statistically significant. With 95 Data Engineer offers in the dataset, the interquartile range runs from **14,875 MAD (Q1) to 22,125 MAD (Q3)**, with a median of 19,000 MAD. The Data Scientist median at Tanger (24,500 MAD) is slightly above the national average — a positive signal for premium hires.

Profile	Tanger Median	Competitive Range (Q2–Q3)
Data Scientist	24,500 MAD	24,500 – 27,500 MAD
Data Engineer	19,000 MAD	19,000 – 22,125 MAD
Data Analyst	13,250 MAD	13,250 – 17,000 MAD
DevOps / SRE	19,000 MAD	19,000 – 21,063 MAD
Cloud Engineer	21,750 MAD	21,750 – 25,000 MAD

To attract competitive candidates, Mexora should position its salary offers between the median and Q3 for each profile. Offers below Q1 risk losing candidates to competitors immediately.

3.4. Question 4 — Does Required Experience Correlate with Proposed Salary?

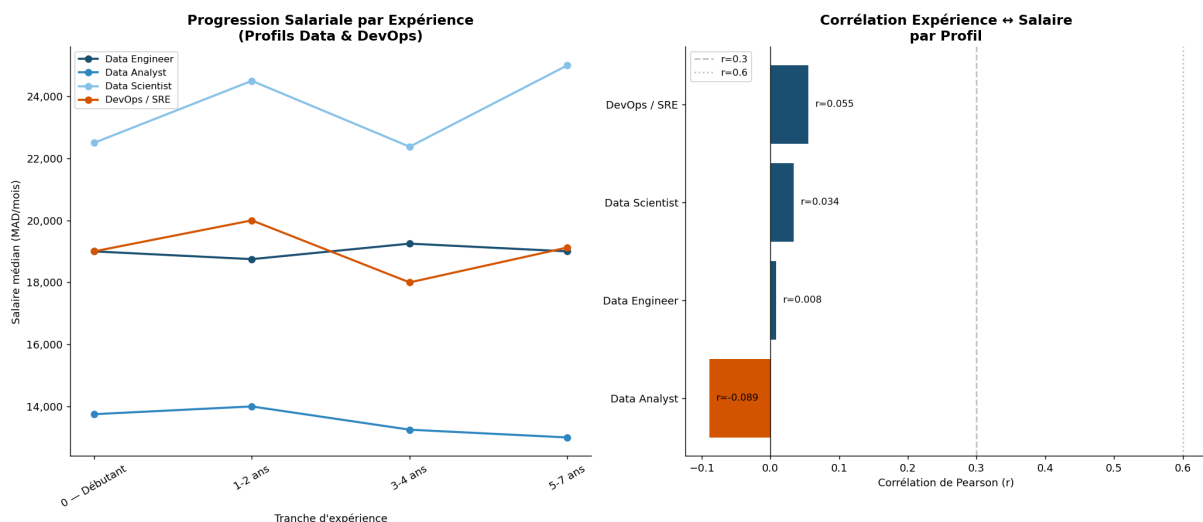


Figure 5. Left: Median salary by experience bracket for four profiles (Data Engineer, Data Analyst, Data Scientist, DevOps/SRE). Right: Pearson correlation coefficient between required experience and proposed salary per profile. Source: `offres_clean.parquet` (Silver zone), direct DuckDB query.

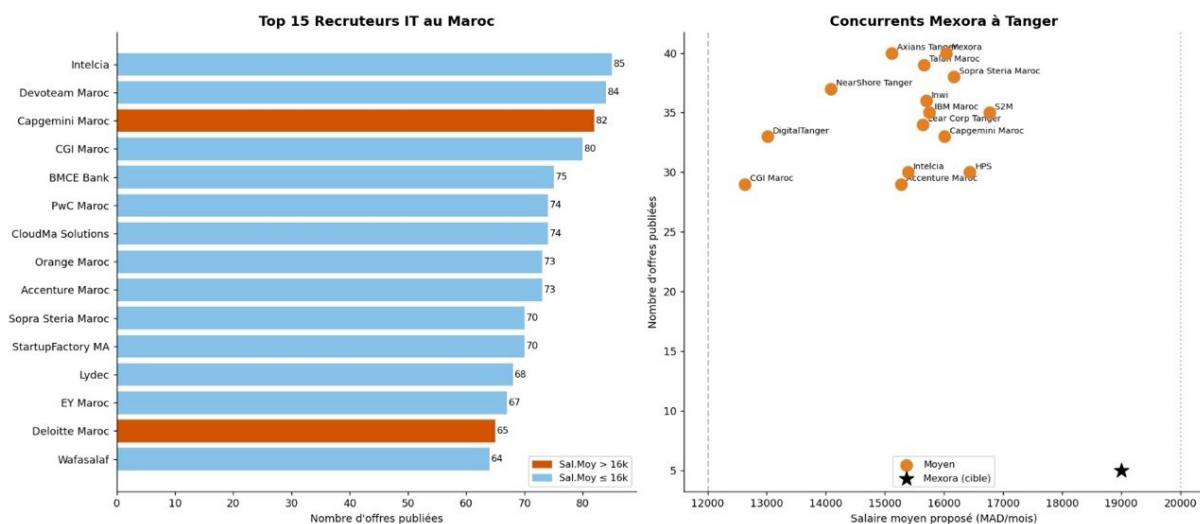
Contrary to the initial hypothesis, the correlation between required experience and proposed salary is **near zero across all profiles analysed** (Data Engineer: $r = 0.008$, Data Scientist: $r = 0.034$, DevOps/SRE: $r = 0.055$, Data Analyst: $r = -0.089$). This is not a data quality issue — it is a genuine structural characteristic of the Moroccan IT labour market.

The salary curves confirm this finding visually: a Data Engineer’s median salary remains remarkably stable between 18,750 MAD (1–2 years) and 19,250 MAD (3–4 years), with no meaningful progression. The same pattern holds for DevOps/SRE profiles. The Data Scientist shows slight upward movement at the senior level but remains below statistical significance.

This pattern likely reflects a market where **talent supply is limited relative to demand**, pushing companies to offer market-rate salaries regardless of seniority in order to attract any qualified candidate. Experience-based premium structuring is not yet standard practice in Morocco's IT market.

This finding gives Mexora strategic flexibility. Rather than applying rigid seniority-based pay scales, Mexora can use a **wide salary band** per profile (e.g., 17,000–25,000 MAD for Data Engineer) and position the actual offer based on the individual candidate's demonstrated technical skill and project portfolio — not purely years of experience. This approach is both market-consistent and allows Mexora to hire junior profiles at competitive rates without sacrificing quality.

3.5. Question 5 — Who Are Mexora's Competitors in the Tanger Talent Market?



recruiter in Tanger — the question is how to strengthen that position for the five priority data hires.

Mexora does not need to match the volume of SSII firms. As a product company, its competitive advantage lies in three areas: (1) **salary above the Tanger median** — positioning at 19,000–22,000 MAD for Data Engineer immediately differentiates from most local SSII competitors; (2) **a genuine data product environment** — candidates who want to work on real data infrastructure, not billable client projects; (3) **hybrid work policy** — Tanger’s 69.5% remote appetite means this is a retention tool, not just an attraction tool.

4. Summary Dashboard

The following section presents the four required dashboard visualisations, combining Power BI and Python/Matplotlib outputs as permitted by the project specification.

4.1. Visual 1 — Geographic Distribution of IT Offers (Morocco Map)

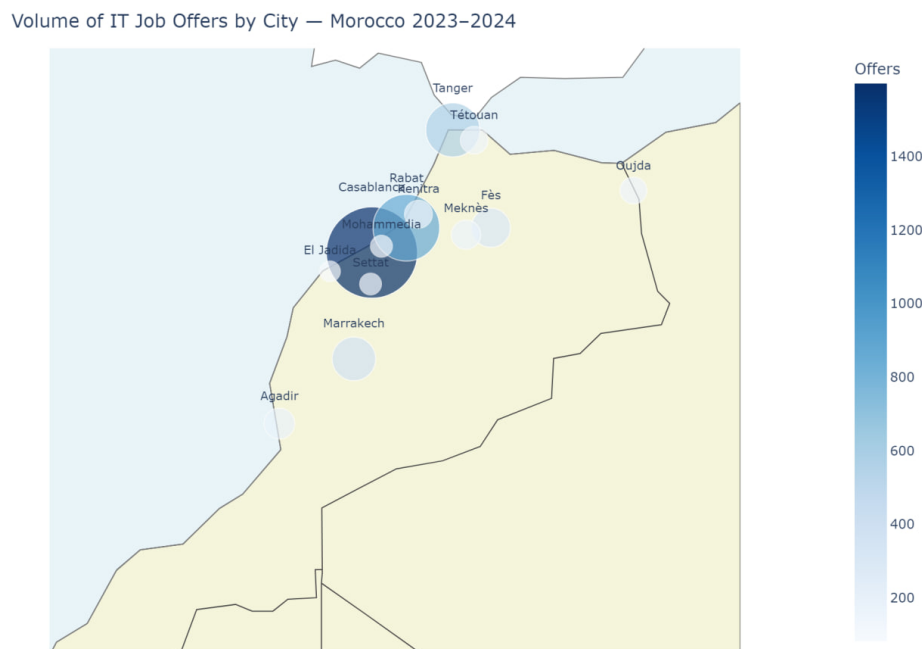


Figure 7. Bubble map of Morocco showing IT job offer volume by city (Jan 2023–Nov 2024). Bubble size and colour intensity are proportional to offer count. Casablanca leads with 1,704 offers; Tanger ranks 3rd with 597. Generated with Plotly/Matplotlib from `offres_par_ville.parquet`.

4.2. Visual 2 — Top 15 IT Skills in Morocco

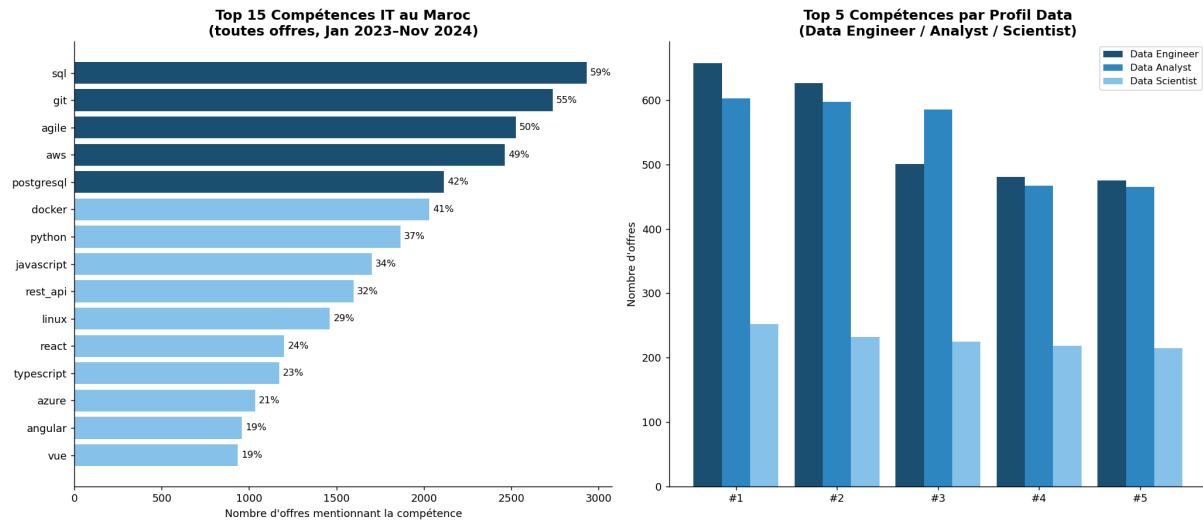


Figure 8. Top 15 IT skills demanded in Morocco across all profiles (left) and top 5 skills per data profile (right). SQL dominates with 58.6% market penetration. Data Engineers uniquely require Spark, ETL, and dbt. Generated from `top_compétences.parquet` (Gold zone).

4.3. Visual 3 — Salary Distribution by Profile (Q1 / Median / Q3)

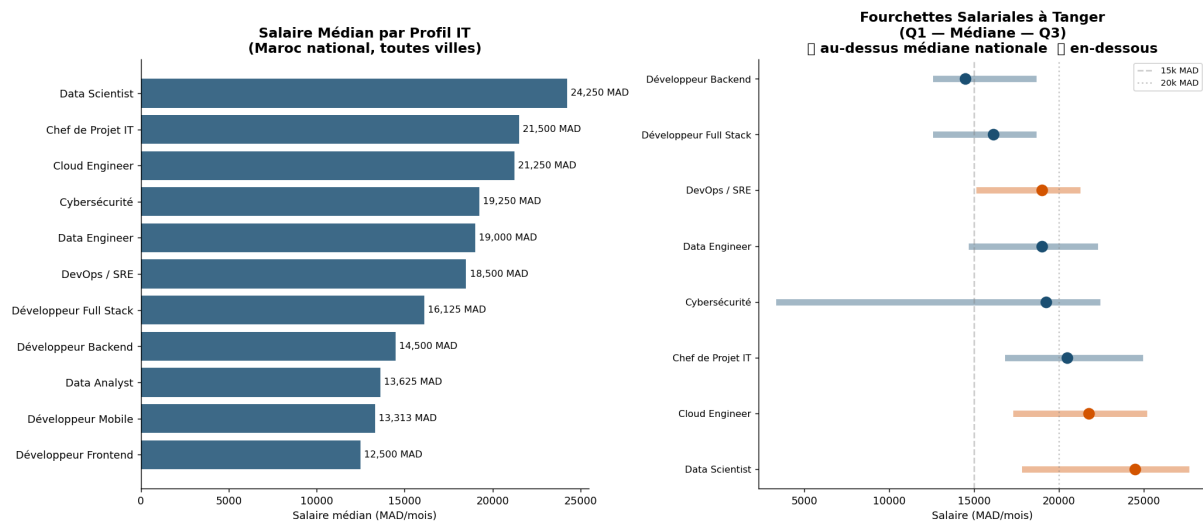


Figure 9. Salary distributions by IT profile. Left: national median salaries. Right: Tanger-specific interquartile ranges (Q1–Median–Q3). Orange profiles are above the national median; blue are at or below. This is the primary salary benchmarking tool for Mexora’s HR Director. Generated from `salaires_par_profil.parquet` (Gold zone).

4.4. Visual 4 — Monthly Evolution of IT Offer Volumes

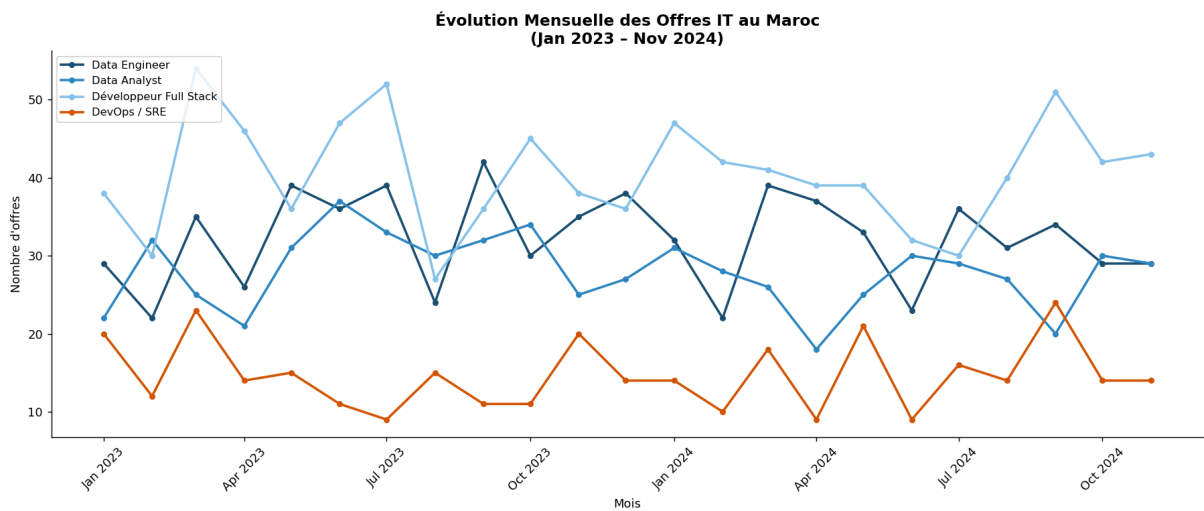


Figure 10. Monthly evolution of IT job offer volumes by profile (Jan 2023–Nov 2024). Développeur Full Stack consistently leads in volume. Data Engineer demand remains stable at 25–40 offers per month with no significant seasonal decline, confirming sustained market demand. Generated from `tendances_mensuelles.parquet` (Gold zone).

4.5. Power BI Dashboard Overview

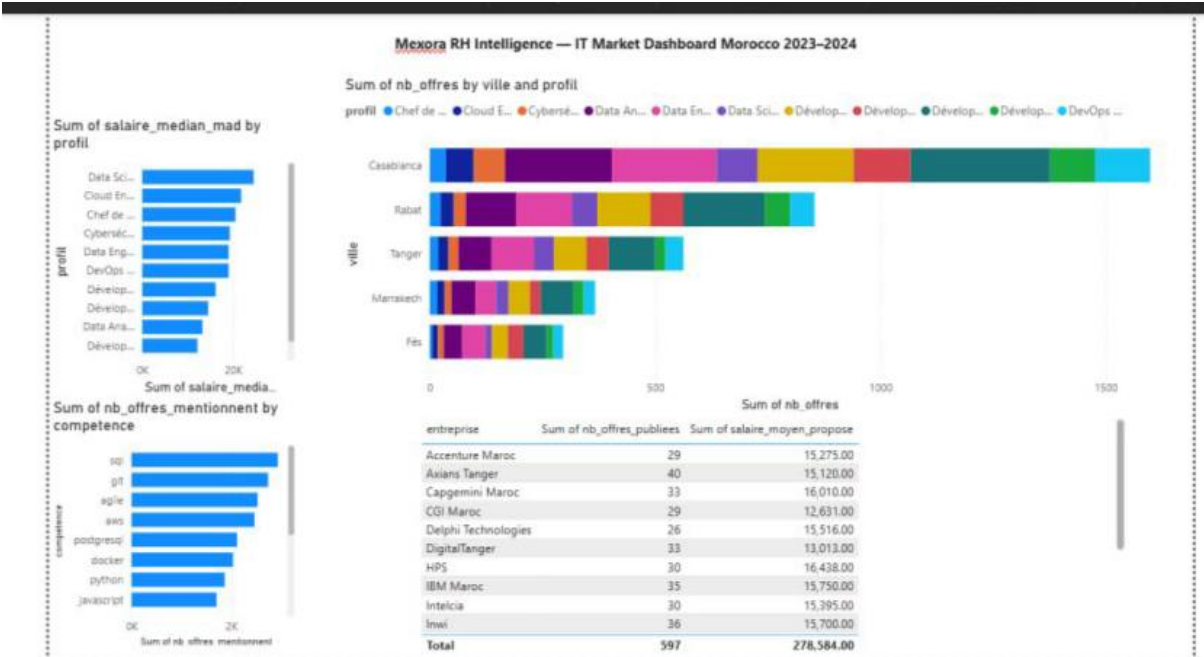


Figure 11. Power BI dashboard — `mexora_Data_lake_dashboard.pbix`. Four interactive visuals: salary by profile (top-left), IT offers by city (top-right), top skills (bottom-left), and Tanger competitor table (bottom-right). Data source: Gold Parquet files connected directly via Power BI Parquet connector.

5. Strategic Recommendations for Mexora

Based on the five analytical questions, the following six concrete recommendations are provided to the HR Director for the five planned data hires.

5.1. Recommended Salary Bands

Profile	Min (MAD)	Max (MAD)	Rationale
Data Engineer	19,000	24,000	Tanger median to Q3
Data Analyst	14,000	18,000	Above Tanger median
Data Scientist	24,000	30,000	Premium profile, scarce supply
DevOps / SRE	18,000	22,000	Tanger median to Q3
Cloud Engineer	21,000	26,000	Competitive vs Tanger Free Zone

5.2. Mandatory Skills Screening

For all five data profiles, the following baseline skills should be verified at first screening: **SQL, Python, Git**. These three skills appear in the top 7 nationally and are non-negotiable in the current market. Beyond the baseline, each profile requires domain-specific verification:

- **Data Engineer:** Spark or PySpark, Airflow or dbt, at least one cloud platform (AWS/GCP/Azure)
- **Data Analyst:** Power BI or Tableau with DAX knowledge, SQL at intermediate level
- **Data Scientist:** scikit-learn, PyTorch or TensorFlow, pandas/numpy, Jupyter
- **DevOps/SRE:** Docker, Kubernetes, CI/CD pipeline experience, Linux administration
- **Cloud Engineer:** Terraform, one primary cloud certification (AWS or Azure), Docker

5.3. Recruitment Channels

- **Rekrute** accounts for 45.4% of Moroccan IT offers — primary posting platform
- **LinkedIn Morocco** reaches senior profiles and passive candidates — essential for Data Scientist and Cloud Engineer
- **MarocAnnonce** covers 29.3% of the market at lower cost — useful for volume roles
- **University partnerships** in Tanger (Universiapolis, ENSA Tanger) for junior Data Engineer and Data Analyst profiles — not captured in this dataset but strategically valuable

5.4. Competitive Differentiation

Mexora's three sustainable competitive advantages over SSII competitors in Tanger are:

1. **Product company positioning:** Candidates work on Mexora's own data infrastructure, not client projects. This is a strong motivator for engineers who want long-term ownership of their work.

2. **Hybrid work policy:** With 69.5% of Tanger’s Data Engineer offers already proposing remote work, a formal 3-days-remote policy is table stakes — not a differentiator. Mexora should offer full flexibility (up to 4 days remote) to stand out.
3. **Salary above the local SSII average:** The analysis shows most Tanger SSII competitors offer 12,500–16,000 MAD on average. A Data Engineer offer at 19,000–22,000 MAD immediately separates Mexora from 70% of the local competition.

6. Conclusion

This study demonstrates that the Moroccan IT job market in 2023–2024 is both active and structurally accessible for Mexora. Tanger is not a secondary market — with 597 IT offers and a 69.5% remote work rate for data profiles, it offers a genuine pool of quality candidates who are actively seeking local opportunities.

The five analytical questions answered in this report give the HR Director precise, data-backed answers to each hiring decision: what skills to require, what salaries to offer, which platforms to use, and which competitors to monitor. The absence of a strong experience-salary correlation means Mexora can recruit on merit and potential rather than being locked into expensive seniority premiums.

The pipeline infrastructure built for this study — 81.9-second end-to-end execution, fully reproducible from a single `python main.py` command — can be rerun as new job offer data becomes available, making it a living market intelligence tool rather than a one-time report.

Mexora should move forward with its five planned data hires. The recommended order of priority, based on market availability and strategic impact, is: (1) Data Engineer, (2) Data Analyst, (3) DevOps/SRE, (4) Data Scientist, (5) Cloud Engineer. The Data Engineer and Data Analyst profiles have the deepest candidate pools in Tanger (95 and 73 offers respectively) and should yield the fastest hires at competitive salary points.

A. Technical Appendix — Data Quality Summary

Anomaly Type	Count	% of Dataset	Action Taken
Incoherent dates (pub > exp)	238	4.8%	Flagged, dates normalised
Unparseable salary	298	5.9%	Flagged, salary set to NULL
Unclassified job title	276	5.5%	Flagged, excluded from Gold
Unrecognised contract type	198	3.9%	Mapped to “Autre”
Unparseable experience	371	7.4%	Flagged, exp set to NULL
Clean offers (0 anomalies)	3,772	75.4%	—

B. Technical Appendix — Gold Tables Schema

Table	Grain	Rows	Key Columns
top_competences.parquet	profil $\times$ skill	201	famille, competence, nb_offres, ran
offres_par_ville.parquet	ville $\times$ profil	143	nb_offres, nb_remote, pct_remote
salaires_par_profil.parquet	profil $\times$ ville	136	median, Q1, Q3, min, max
entreprises_recruteurs.parquet	entreprise $\times$ ville	323	nb_offres, sal_moyen, profils[]
tendances_mensuelles.parquet	year $\times$ month $\times$ profil	253	nb_offres, sal_moyen_mois

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Full pipeline source code: github.com/Matthews-Lungu/Mexora_rh_Data_lake